EpiGen

Protease-gated selective antibiotics, with agentic compensatory-mutation design

re:AGENT hackathon — August 15, 2026

The problem

- Broad-spectrum antibiotic proteins kill **everything** — pathogens *and* commensal ("good") bacteria.
- Killing commensals has real downstream cost: dysbiosis, infection susceptibility, long antibiotic-course side effects.
- We want a drug that's still fully active against pathogens, but **locally silenced** wherever a healthy microbiome already is.

The idea: a protease-gated switch

- Commensal *Lactobacillus* spp. secrete a protease (lactocepin, PrtP) that pathogens don't.
- Engineer lactocepin's substrate motif into a surface-exposed loop of the antibiotic protein.
- Near commensals: local protease cleaves the motif, inactivates the drug. Everywhere else: drug stays fully active.

The catch

Inserting a foreign motif destabilizes the scaffold. We need **compensatory mutations** that restore structural stability *without* burying the motif (it has to stay cleavable).

Demo case

- **Scaffold:** hen egg-white lysozyme — 129 aa, well-characterized, fast-folding, real PDB structure (193L).
- **Insertion:** lactocepin substrate motif, in a surface-exposed loop.
- **Goal:** pathogen-killing activity preserved; activity locally silenced near *Lactobacillus*.

[screenshot: structure viewer, WT with edit loop highlighted]

Pipeline: fold → invert → refold → search

1. Fold the edited sequence (ESMFold2), gate on pLDDT/PAE.
2. Inverse-fold (ProteinMPNN) diverse warm starts; MCMC then proposes *uniform-random* single-residue moves over the whole scaffold minus the fixed edit — the sampler itself is blind, not model-guided.
3. Accept/reject every proposal by a combined score from three independent experts: **ESM2** (sequence LM), **ProteinMPNN** (structure-conditioned), **Evo2** (DNA-level, causal) — this is where the models actually steer the search.
4. Refold top candidates, self-consistency (TM-score) gate before trusting anything downstream.

Three legs of evidence, per candidate

Contact/energy

Per-neighbor Δdistance, Δcontact energy, ΔpLDDT/PAE near every changed residue.

SAE interpretability

ESM-C sparse-autoencoder feature deltas across WT / edit-only / compensated — top-k surfaced, not a full sweep.

Agent explanation

Grounded in the numeric deltas above — flags itself if a claimed rescue contradicts the numbers.

All three feed one plain-language explanation of *whether and how* a compensatory mutation rescues stability.

Live: candidate scores vs. WT

[screenshot: Results page,
chain starting/ending scores histogram + WT baseline]

Live: structure & agent explanation

[screenshot: Structure viewer,
top candidate colored by SAE ΔΔ, agent explanation panel]

What's real vs. what's next

Working end to end

- Fold/invert/refold + TM gate
- 3-expert MCMC search (ESM2/PMPNN/Evo2)
- Contact-energy + SAE diffs
- Grounded agent explanation
- Literature-annotation conflict check

Next

- Wet-lab validation of motif accessibility
- Insertion (not just substitution) edits
- Oligo library export for synthesis
- Broader scaffold library beyond lysozyme

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A selective antibiotic, and the agent that explains why it should still fold.

Questions?